HACKHARVARD 2026 · REGENERON: REINVENT CLINICAL TRIALS & DRUG DEVELOPMENT · CHALK



ProtocolOS

Every exclusion deserves a rationale.

A rationale reviewer for oncology eligibility criteria: it checks why each lab, ECOG and washout limit is there, against earlier comparable trials and FDA final guidance, and a deterministic verifier checks every suggested change.

protocolos-web.vercel.app · github.com/rinatrizvanov/ProtocolOS · formerly TrialLens

Research prototype · not clinical or regulatory advice

MIT licensed

THE BOTTLENECK

Criteria are carried forward from precedent, without a documented rationale

- Lab, ECOG and washout limits are often copied from the last protocol. Nobody writes down **why** each one is there, so nobody can tell which ones still have a reason.
- **57%** of protocols have at least one substantial amendment; **45%** of those amendments were judged avoidable (Getz et al. 2016).
- Protocol teams rarely get evidence-based feedback on a draft until it is too late to change cheaply (Regeneron, Trial by Design Pt 2).
- **FDA final guidance (July 2026)** covers exactly these criteria: lab values, performance status, washout periods. For older trials we apply it *retrospectively* and label it so.

> *"Unnecessarily restrictive eligibility criteria may slow subject accrual, limit patients' access to clinical trials, and lead to trial results that do not fully represent treatment effects..."*

FDA final guidance, Cancer Clinical Trial Eligibility Criteria: Laboratory Values (July 2026), §I

But "loosen the criteria" misses the point

Restrictions have reasons: patient safety with myelosuppressive chemotherapy, enrichment, interpretable adverse events. The question is whether each limit's rationale is documented and still holds.

TWO ENTRY MODES

One engine, two points in the protocol's life

A · Finishing a trial

Review an existing protocol before it is finalized or amended

- Reads the posted protocol PDF: each criterion with its verbatim line and page
- **Rationale** badge per criterion, with the arm whose drugs drive it
- Compare with earlier comparable trials; Ask; **Redraft** with live checks
- **Decide** with structured reasons; export a review packet

B · Create criteria from an idea

Draft criteria for a new concept: condition, drugs, setting, phase

- Finds precedent trials that started before the design date
- Drafts each criterion from a **verified** precedent value or FDA passage
- No verified source: `Insufficient evidence`, never a guess
- Self-critique of every draft; before/after against a real protocol

Protocol or idea
NCT ID, posted PDF, or a concept

→

Rationale
precedents · FDA final guidance · regimen

→

Deterministic verifier
the authority on every value

→

The team decides
reasons recorded in the packet

Users: protocol authors, clinical scientists, protocol-review teams. ProtocolOS prepares the rationale; the team decides.

MODE A · FINISHING A TRIAL

Each criterion gets a rationale, a comparison and a checked redraft

Rationale badges

Review recommended

Rationale supports keeping

Insufficient evidence

Likely purpose (safety, efficacy enrichment, regulatory precedent), *inferred* unless the protocol states it, and an **arm line**: which arm's drugs are linked to the toxicity.

Compare and Ask

Side by side with comparable trials that **started earlier**, with regimen coverage and page links. Ask answers questions from the provenance graph; every statement is checked against it.

Redraft workspace

Type another limit and the **deterministic checks run live**: precedent, FDA passage, regimen logic. Outcomes: Supported · Qualified · Rejected · **Needs medical review**.

Verify

Challenge any flag. A plausible argument with a true premise can still be rejected when another arm contradicts it.

Decide

Structured reasons: safety concern, regulatory requirement, efficacy enrichment, prior evidence, operational reason; free text only for exceptions.

Review packet

Verbatim requirement and page, rationale for and against, open questions, next action and the team's decision. Markdown and JSON.

Hemoglobin ≥10 g/dL: a review question, not a mistake

SIDE-BY-SIDE WITH COMPARABLE TRIALS

TRIAL	REGIMEN	THRESHOLD	VS TARGET	REGIMEN COVERAGE	SOURCE
NCT03409614 (reviewed)	**cemiplimab, ipilimumab, pemetrexed, carboplatin, cisplatin, paclitaxel**	**≥10 g/dL**	—	—	p.67
CheckMate 9LA NCT03215706 · Bristol-Myers Squibb · start 2017-08-24	carboplatin, cisplatin, ipilimumab, nivolumab, paclitaxel, pemetrexed	≥9 g/dL	comparator looser	full	p.42
eNERGY NCT03351361 · Rennes University Hospital · start 2018-02-19	carboplatin, ipilimumab, nivolumab, paclitaxel, pemetrexed	>10 g/dL	same	full	p.8
CheckMate 227 NCT02477826 · Bristol-Myers Squibb · start 2015-08-05	carboplatin, cisplatin, gemcitabine, ipilimumab, nivolumab, paclitaxel, pemetrexed	≥9 g/dL	comparator looser	full	p.73
NCT02220894 NCT02220894 · Merck Sharp & Dohme LLC · start 2014-10-30	carboplatin, paclitaxel, pembrolizumab, pemetrexed	≥9 g/dL	comparator looser	full	p.33

Coverage = whether the comparator's regimen contains the same toxicity-relevant drug classes as this trial (e.g. myelosuppressive classes for hematologic criteria). Partial coverage weakens the comparison.

- **Own rationale first:** the arms include platinum doublets with pemetrexed or paclitaxel; the protocol notes higher grade 3–4 anemia with pemetrexed
- A safety rationale for *a* threshold exists; the precise value is not established by the available evidence
- **Earlier peers:** CheckMate 9LA, CheckMate 227 and KEYNOTE-042 used **≥9 g/dL**; eNERGY used **>10 g/dL** → **Review recommended**
- **Verify:** "cemiplimab is not myelosuppressive, so lower the ANC floor" is **Rejected**: other arms contain carboplatin, cisplatin, paclitaxel, pemetrexed
- Not claimed: that 10 is wrong, or that a lower limit is safe

MODE B · CREATE CRITERIA FROM AN IDEA

Verifier-gated drafts, then a before/after against EMPOWER-Lung 3 as written

Idea: first-line metastatic NSCLC; cemiplimab, ipilimumab, platinum doublet with pemetrexed or paclitaxel; phase 3; design date March 2018.

- Precedents limited to trials that started **before** the design date
- A value is drafted only when the verifier confirms it in a precedent PDF or an FDA passage; otherwise **Insufficient evidence**
- **Self-critique:** each draft is reviewed by the same engine that reviews finished protocols
- Differences from the real protocol are review questions, not errors

Guided demo: protocolos-web.vercel.app/tour/draft

CRITERION	DRAFTED	SOURCE	EL3 AS WRITTEN
Hemoglobin	**≥9.0 g/dL**	CheckMate 9LA	≥10.0 g/dL
Creatinine clearance	**≥50 mL/min**	CheckMate 9LA	>30 mL/min
Investigational washout	**≥28 days**	KEYNOTE-042	≥30 days
Surgery washout	**≥21 days**	KEYNOTE-042	≥28 days
ANC	≥1.5 ×10⁹/L	CheckMate 9LA	≥1.5 ×10⁹/L
Platelets	≥100 ×10⁹/L	CheckMate 9LA	≥100 ×10⁹/L
Bilirubin · AST/ALT	≤1.5 · ≤3.0 ×ULN	CheckMate 9LA	≤1.5 · ≤3.0 ×ULN
ECOG	≤1	CheckMate 9LA	≤1

From evals/draft/results.json (NCT03409614). A drafted value shows a precedent, not that it is safe.

MEASURED, NOT PREDICTED

What we measured, on committed data

0

criteria drafted without a verified precedent across the draft eval on 4 real trials; no overclaiming wording found. Drafts agree with the protocol as written for **22/31** comparable criteria.

336/336

planted investigator defects rejected (unknown trial, changed number, causal phrase, one-word misquote, dropped counter-evidence, injected instruction); **44/44** verbatim controls kept.

24/24

verifier faults rejected and **0/10** valid changes rejected, in a Chalk Evaluation over 39 cases.

122/128

criteria extracted from 19 real protocol PDFs (97.6% precision), each with its verbatim line and page.

24/24

planted stricter thresholds never marked as supported; **0/10** loosened thresholds falsely flagged (synthetic edits of authentic protocols).

NHANES

Share of US adults below the EL3 hemoglobin limit and below the ≥9 g/dL peer limit, survey-weighted (NHANES 2017-2018). **Caveat:** a US population share, not a trial-eligible share and not an enrollment prediction.

Developers wrote both the planted defects and the checks; single annotator; no clinician review yet. Scripts in evals/ reproduce every number.

READINESS

Deployed, documented, installable

Use it now

- **App:** protocolos-web.vercel.app (`/review`, `/draft`, `/tour`, `/tour/draft`)
- **API:** protocolos-api-production.up.railway.app, OpenAPI page at `/docs`
- **Code:** github.com/rinatrizvanov/ProtocolOS, MIT licensed
- Public demo runs the deterministic engine; no model output is substituted when no model key is configured

Install and integrate

- Python + FastAPI engine; Next.js app; one Dockerfile
- `docs/INSTALL.md`: clean-clone install, prewarm, run both UIs
- 164 automated tests pass; end-to-end demo check and a numbers check gate every pitch number
- Structured JSON over HTTP: plugs into a protocol-review workflow; packet export as Markdown and JSON

Traceable
every value links to a page of a public document

Reproducible
committed eval results, scripts in `evals/`

Auditable
verifier checks and investigator audit log

AGENT RISKS AND MITIGATIONS

The deterministic verifier is the authority, not the model

RISK	MITIGATION
A model invents a value, a quote or a trial	Every statement is checked against the provenance graph; numbers, quotes and trial IDs must match. The model may only **downgrade** a flag, never upgrade it
A drafted criterion has no real basis	Drafts need a verified precedent or FDA passage; otherwise **Insufficient evidence**
Agreement mistaken for evidence	Agreement among comparators may be template carry-forward; a looser peer shows use, not safety
Instructions hidden in a PDF	Document text is data, never instructions; injected-instruction defects are in the eval
Causal or outcome overclaims	No claims of safety, enrollment or cost gains, or causality between trials; wording checks in tests
Patient privacy	Public data only: ClinicalTrials.gov, posted protocols, FDA guidance, NHANES. No patient data
Blind spot	Rationale in non-public sources (preclinical toxicology, internal development data, regulator correspondence) cannot be seen by ProtocolOS

BUILT ON CHALK

Chalk does real work in ProtocolOS

Features + resolver

A `Trial` feature set resolved live from ClinicalTrials.gov inside Chalk (query `triallens_registry_profile`). It answers per trial: is the threshold even in the registry? For EMPOWER-Lung 3 the lab criteria are only in the protocol PDF.

Compute + Evaluations

The verifier runs as a Chalk function over 39 cases graded by Chalk scorers: **run succeeded, 0 task errors**, 24/24 faults rejected. The `protocolos-investigator` suite has run locally.

Model Gateway

Router-backed critic and Ask answers. The model may only downgrade, and every statement passes the verifier. The public demo stays deterministic.

Verifier evaluation `cmvco3m0n01dw01y520oc80sq` · environment `trllf247b678` · Chalk team `triallens`

SCOPE AND WHAT'S NEXT

What ProtocolOS is, and isn't

Is

- A working, deployed rationale reviewer and drafter for lab, ECOG and washout criteria in oncology
- Grounded in public protocols and FDA final guidance (July 2026), with a deterministic verifier
- MIT licensed, documented install, reproducible evals

Isn't (yet)

- Clinically validated: rule-based, single-annotator gold, no clinician review
- A recruitment, cost or safety predictor; we make no such claims
- **Next:** internal precedent and amendments · ICH M11 structured protocols · clinician review

Every exclusion deserves a rationale.

Try it: protocolos-web.vercel.app · Code: github.com/rinatrizvanov/ProtocolOS